Practice Problem D
Squeeze the Cylinders
Time limit: 2 seconds

Laid on the flat ground in the stockyard are n heavy metal cylinders with possibly different diameters but with the same length. Their ends are aligned and their axes are oriented to exactly the same direction.

We'd like to minimize the area occupied. The cylinders are too heavy to lift up, although rolling them is not too difficult. So, we decided to push the cylinders with two high walls from both sides.

Your task is to compute the minimum possible distance between the two walls when cylinders are squeezed as much as possible. Cylinders and walls may touch one another. They cannot be lifted up from the ground, and thus their order cannot be altered.

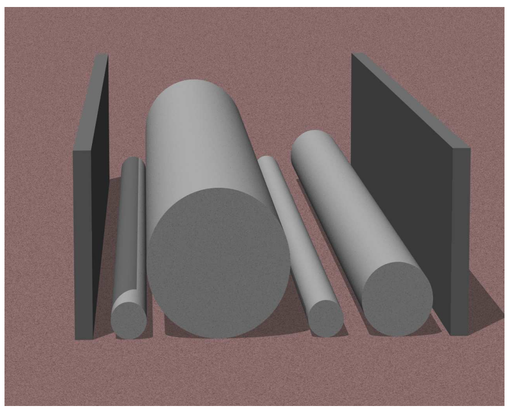

Figure D.1: Cylinders between two walls.

Input

The first line of input contains one integer n ($1 \leq n \leq 500$). The second line contains n positive integers not more than $10\,000$, representing the radii of cylinders from left to right.

Output

Output the distance between the two walls when they fully squeeze up the cylinders. The relative error of the output must be within 10^{-9}.

Sample Input #1	Sample Output #1
2 10 10	40.000000000

Sample Input #2	Sample Output #2
2 4 12	29.856406461

Sample Input #3	Sample Output #3
5 1 10 1 10 1	40.000000000

Explanation for the sample input/output #1

Figure D.2 illustrates possible positions of the cylinders to achieve the minimum possible distance between the two walls in the sample inputs.

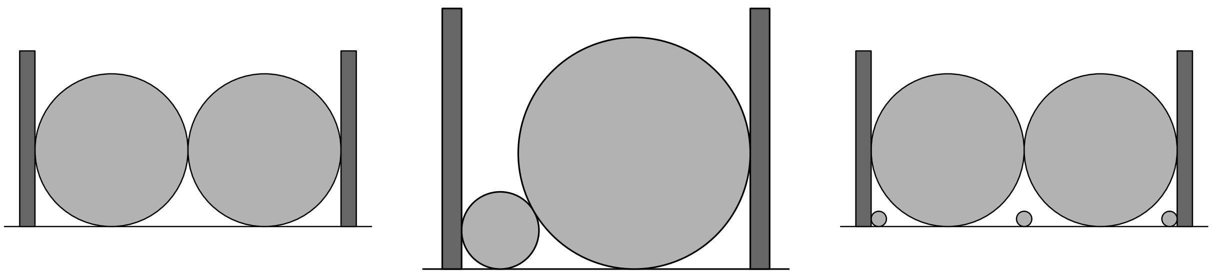

Figure D.2: Illustrations of the sample inputs (from left to right).

(Note: This task has been adapted from ICPC Asia Tsukuba Regional Contest 2015 with an approval from the contest judge)